

Design of Multiple Ontology Based Agro Knowledge Mining Model

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ABSTRACT ORIGINAL

Farming is regarded as a major industry in India, accounting for 17% of the country's GDP growth. Agriculture employs 60% of the population hence it is considered an important sector in India. The important factors for agriculture are pest management, disease prevention, irrigation management, soil mineral composition, crop management, location, and the season in which the crop is grown. Hence all this information along with the techniques are well known only by the experienced farmers. Hence it is important to create an agro knowledge management system. As a result, this work makes an attempt to develop a multiple ontology-based agro knowledge management system. The designed system consists of agriculture information related to attributes of soil mineral, moisture, season, location, crop type, and temperature. It consists of multiple ontologies such as soil ontology, crop ontology, location ontology, and crop season ontology to provide agronomy knowledge. Soil ontology is premeditated to classify the soil type in a hierarchical order while crop ontology classifies the crop type, location ontology classifies locations suitable for different crop types and finally, crop season ontology classifies the season that is suitable for different crops. A rule base is built to develop the knowledge base and to validate the truthfulness of the knowledge base. Visualization of a knowledge base is carried out for better understanding and decision-making. © 2023 International Journal on Recent and Innovation Trends in Computing and Communication. All rights reserved.